WEEK-7

Aim

To execute and understand SQL queries on an e-commerce database, including retrieving records, filtering and sorting data, joining related tables, and calculating summary counts.

Learning Objectives

- Retrieve records from customer, product, order, inventory, order-detail, and review-related tables.
- Filter records by status, category, price, rating, date, and stock quantity using WHERE conditions.
- Sort results with ORDER BY in ascending and descending order.
- Combine related data using JOIN operations.
- Use COUNT(*) and a subquery to analyze records and prices.

1 Display customer records

SELECT * FROM CUSTOMER displays all columns and rows in CUSTOMER, allowing customer IDs, names, contact details, addresses, and registration dates to be checked.

```
SQL> SELECT * FROM CUSTOMER;

CUSTOMER_ID C_NAME          EMAIL                          PHONE_NO ADDRESS                 REG_DATE
----------- --------------- ------------------------------ ---------- ----------------------- ------------
        201 DEVI            devi@gmail.com                 9000000001 Chennai                 22-OCT-26
        202 DIVI            divi@gmail.com                 9000000002 Chengalpattu            22-OCT-26
        203 KAVYA           kavya@gmail.com                9000000003 Tambaram                22-OCT-26
        101 Arun            arun@gmail.com                 9876543210 Chennai                 19-OCT-26
        102 Priya           priya@gmail.com                9876543211 Madurai                 19-OCT-26
        103 Kumar           kumar@gmail.com                9876543212 Coimbatore              19-OCT-26
        104 Divya           divya@gmail.com                9876543213 Trichy                  19-OCT-26
        105 Karthik         karthik@gmail.com              9876543214 Salem                   19-OCT-26

8 rows selected.
```

2 Display product records

SELECT * FROM PRODUCT retrieves the product catalogue, including product IDs, names, prices, categories, and descriptions.

+

```
SQL> SELECT * FROM PRODUCT;

PRODUCT_ID PRODUCT_NAME               PRICE CATEGORY            DESCRIPTION
---------- ------------------------ --------- ------------------- ----------------------------------------
       101  RESIN KEYCHAIN            250.00 KEYCHAIN            HANDMADE RESIN KEYCHAIN
       102 RESIN BOOKMARK             200.00 STATIONARY          HANDMADE RESIN BOOKMARK
       103 Resin Pendant              350.00 Jewellery           Handmade resin pendant
       206 Resin Art Keychain         150.00 Resin Art           Customized handmade resin keychain
       207 Resin Art Coaster Set      450.00 Resin Art           Handmade decorative resin coasters
       208 Resin Art Pendant          250.00 Resin Art           Floral resin pendant necklace
       209 Resin Art Photo Frame      600.00 Resin Art           Customized resin photo frame
       210 Resin Art Bookmark         180.00 Resin Art           Pressed flower resin bookmark

8 rows selected.
```

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM PRODUCT;

PRODUCT_NAME                    PRICE
------------------------ ---------
 RESIN KEYCHAIN                250.00
RESIN BOOKMARK                 200.00
Resin Pendant                  350.00
Resin Art Keychain             150.00
Resin Art Coaster Set          450.00
Resin Art Pendant              250.00
Resin Art Photo Frame          600.00
Resin Art Bookmark             180.00

8 rows selected.
```

4 Display all orders

SELECT * FROM ORDERS displays order IDs, customer IDs, order dates, total amounts, and order statuses.

```
SQL> SELECT * FROM ORDERS;

ORDER_ID CUSTOMER ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------- -------- --------- ------------ -------------------------
0201     C201     22-AUG-26          750 Pending
0202     C202     23-AUG-26         1000 Shipped
0203     C203     24-AUG-26          500 Delivered
0204     C204     25-AUG-26          750 Pending
0205     C205     26-AUG-26         1000 Shipped
0206     C206     27-AUG-26         1200 Shipped
0207     C207     28-AUG-26          700 Delivered
0208     C208     29-AUG-26         1900 Delivered

8 rows selected.

SQL> COMMIT;

Commit complete.
```

```
SQL> SELECT *
  2  FROM PAYMENT
  3  ORDER BY PAYMENT_ID ASC;

PAYMENT_ID ORDER_ID PRODUCT_ID PRODUCT_NAME              PAYMENT_D PAYMENT_MODE       PAYMENT_STATUS            TRANSACTION_AMOUNT
---------- -------- ---------- ------------------------ --------- ------------------ ------------------------ ------------------
         1 0201     101        RESIN KEYCHAIN           22-AUG-26 UPI Pending                          750
         2 0202     102        RESIN ART                23-AUG-26 Credit Card        Completed                             1000
         3 0203     103        RESIN JEWELRY            24-AUG-26 Debit Card         Completed                              500
         4 0204     104        HOME DECOR               25-AUG-26 Net Banking        Completed                             7000
         5 0205     105        RESIN NECKLACE           26-AUG-26 UPI Completed                                    5200
         6 0206     106        RESIN BRACELET           27-AUG-26 Credit Card        Completed                             2450
         7 0207     107        RESIN EARRINGS           28-AUG-26 Debit Card         Completed                            20000
         8 0208     108        RESIN WALL ART           29-AUG-26 Debit Card         Failed                               26000

8 rows selected.
```

6 Filter products by price

The query applies a price condition to PRODUCT and displays only records matching that condition.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE > 5000;

no rows selected

SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE < 5000;

PRODUCT_ID PRODUCT_NAME                 PRICE CATEGORY             DESCRIPTION
---------- ------------------------ --------- -------------------- ----------------------------------------
       101  RESIN KEYCHAIN            250.00 KEYCHAIN             HANDMADE RESIN KEYCHAIN
       102 RESIN BOOKMARK             200.00 STATIONARY           HANDMADE RESIN BOOKMARK
       103 Resin Pendant              350.00 Jewellery            Handmade resin pendant
       206 Resin Art Keychain         150.00 Resin Art            Customized handmade resin keychain
       207 Resin Art Coaster Set      450.00 Resin Art            Handmade decorative resin coasters
       208 Resin Art Pendant          250.00 Resin Art            Floral resin pendant necklace
       209 Resin Art Photo Frame      600.00 Resin Art            Customized resin photo frame
       210 Resin Art Bookmark         180.00 Resin Art            Pressed flower resin bookmark

8 rows selected.
```

```
SQL> SELECT * FROM INVENTORY;

INVENTORY_ PRODUCT_ID           SELLER_ID            STOCK_QUATITY STOCK
_STATUS         LAST_UPDATED
---------- -------------------- -------------------- ------------- -----
-------------- --------------------
Inv0001    P100001              S00001                         4 2026-
09-12           12-SEP-26
Inv0002    P100001              S00002                         4 Avail
able            13-SEP-26
Inv0003    P100003              S00003                        10 Avail
able            14-SEP-26
Inv0004    P100004              S00004                        15 Avail
able            15-SEP-26
Inv0005    P100005              S00005                         8 Avail
able            16-SEP-26
Inv0006    P100006              S00006                        12 Avail
able            17-SEP-26
Inv0007    P100007              S00007                        18 Avail
able            18-SEP-26
Inv0008    P100008              S00008                        20 Avail
able            19-SEP-26

8 rows selected.
```

8 Filter inventory by availability

The query selects inventory rows whose STOCK_STATUS is 'Available', helping identify items marked as available

```
SQL>  SELECT * FROM INVENTORY WHERE STOCK_STATUS = 'Available';

INVENTORY_ PRODUCT_ID           SELLER_ID            STOCK_QUATITY STOCK_STATUS         LAST_UPDATED
---------- -------------------- -------------------- ------------- -------------------- --------------------
Inv0002    P100001              S00002                         4 Available            13-SEP-26
Inv0003    P100003              S00003                        10 Available            14-SEP-26
Inv0004    P100004              S00004                        15 Available            15-SEP-26
Inv0005    P100005              S00005                         8 Available            16-SEP-26
Inv0006    P100006              S00006                        12 Available            17-SEP-26
Inv0007    P100007              S00007                        18 Available            18-SEP-26
Inv0008    P100008              S00008                        20 Available            19-SEP-26

7 rows selected.
```

```
SQL> SELECT * FROM ORDERS WHERE ORDER_STATUS = 'Delivered';

ORDER_ID CUSTOMER ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------- -------- --------- ------------ -------------------------
0203     C203     24-AUG-26          500 Delivered
0207     C207     28-AUG-26          700 Delivered
0208     C208     29-AUG-26         1900 Delivered
```

10 Display reviews with ratings above 4

The query filters review records using RATING > 4 and displays the matching review details.

```
SQL> SELECT *
  2  FROM REVIEW
  3  WHERE RATING > 4;

REVIEWI CUSTOMERID  PRODUCTID     RATING REVIEWTEXT
   REVIEWDAT
------- ---------- ---------- ---------- ------------------------------
------------------ ---------
R000004        201        206          5 Excellent resin art quality
   15-SEP-26
R000008        105        208          5 Excellent pendant design
   18-SEP-26
R000010        202        210          5 Amazing resin bookmark
   20-OCT-26
```

11 Sort products by ascending price

ORDER BY PRICE ASC sorts products from the lowest price to the highest price.

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  ORDER BY PRICE ASC;

PRODUCT_NAME                    PRICE
----------------------- ----------
Resin Art Keychain             150.00
Resin Art Bookmark             180.00
RESIN BOOKMARK                 200.00
Resin Art Pendant              250.00
 RESIN KEYCHAIN                250.00
Resin Pendant                  350.00
Resin Art Coaster Set          450.00
Resin Art Photo Frame          600.00

8 rows selected.
```

12 Display customer records in order

```
SQL> SELECT *
  2  FROM CUSTOMER
  3  ORDER BY C_NAME ASC;

CUSTOMER_ID C_NAME          EMAIL                          PHONE_NO
 ADDRESS                   REG_DATE
---------- --------------- ------------------------------ ----------
 ------------------------ ------------
       101 Arun             arun@gmail.com                 9876543210
 Chennai                   19-OCT-26
       201 DEVI             devi@gmail.com                 9000000001
 Chennai                   22-OCT-26
       202 DIVI             divi@gmail.com                 9000000002
 Chengalpattu              22-OCT-26
       104 Divya            divya@gmail.com                9876543213
 Trichy                    19-OCT-26
       203 KAVYA            kavya@gmail.com                9000000003
 Tambaram                  22-OCT-26
       105 Karthik          karthik@gmail.com              9876543214
 Salem                     19-OCT-26
       103 Kumar            kumar@gmail.com                9876543212
 Coimbatore                19-OCT-26
       102 Priya            priya@gmail.com                9876543211
 Madurai                   19-OCT-26

8 rows selected.
```

13 Sort products by descending price

ORDER BY PRICE DESC sorts products from the highest price to the lowest price.

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM PRODUCT
  3  ORDER BY PRICE DESC;

PRODUCT_NAME                    PRICE
------------------------- ----------
Resin Art Photo Frame          600.00
Resin Art Coaster Set          450.00
Resin Pendant                  350.00
 RESIN KEYCHAIN                250.00
Resin Art Pendant              250.00
RESIN BOOKMARK                 200.00
Resin Art Bookmark             180.00
Resin Art Keychain             150.00

8 rows selected.
```

14 Sort orders by latest date

ORDER BY ORDER_DATE DESC lists the newest orders first, followed by older orders.

```
SQL> SELECT *
  2  FROM ORDERS
  3  ORDER BY ORDER_DATE DESC;

ORDER_ID CUSTOMER ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------- -------- --------- ------------ --------------------
0208     C208     29-AUG-26         1900 Delivered
0207     C207     28-AUG-26          700 Delivered
0206     C206     27-AUG-26         1200 Shipped
0205     C205     26-AUG-26         1000 Shipped
0204     C204     25-AUG-26          750 Pending
0203     C203     24-AUG-26          500 Delivered
0202     C202     23-AUG-26         1000 Shipped
0201     C201     22-AUG-26          750 Pending

8 rows selected.
```

15 Filter products by category and price

This query filters PRODUCT records using the category and price condition shown in the SQL statement.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE BETWEEN 200 AND 400;

PRODUCT_ID PRODUCT_NAME                PRICE CATEGORY            DESCRIPTION
---------- ------------------------ -------- ------------------ ------------------------------------
       101  RESIN KEYCHAIN            250.00 KEYCHAIN           HANDMADE RESIN KEYCHAIN
       102 RESIN BOOKMARK             200.00 STATIONARY         HANDMADE RESIN BOOKMARK
       103 Resin Pendant              350.00 Jewellery          Handmade resin pendant
       208 Resin Art Pendant          250.00 Resin Art          Floral resin pendant necklace
```

16 Find products in the Keychain category

The query selects products where CATEGORY = 'KEYCHAIN', narrowing the results to that category.

```
SQL> SELECT * FROM PRODUCT WHERE CATEGORY = 'KEYCHAIN';

PRODUCT_ID PRODUCT_NAME                PRICE CATEGORY            DESCRIPTION
---------- ------------------------ -------- ------------------ ------------------------
       101  RESIN KEYCHAIN            250.00 KEYCHAIN           HANDMADE RESIN KEYCHAIN
```

17 Find inventory with higher stock

The query filters INVENTORY using a STOCK_QUANTITY threshold to identify records with higher stock levels.

```
SQL> SELECT *
  2  FROM INVENTORY
  3  WHERE STOCK_QUATITY < 20;

INVENTORY_ PRODUCT_ID           SELLER_ID           STOCK_QUATITY STOCK
_STATUS         LAST_UPDATED
---------- -------------------- -------------------- ------------- -----
-------------- --------------------
Inv0001    P100001              S00001                         4 2026-
09-12          12-SEP-26
Inv0002    P100001              S00002                         4 Avail
able           13-SEP-26
Inv0003    P100003              S00003                        10 Avail
able           14-SEP-26
Inv0004    P100004              S00004                        15 Avail
able           15-SEP-26
Inv0005    P100005              S00005                         8 Avail
able           16-SEP-26
Inv0006    P100006              S00006                        12 Avail
able           17-SEP-26
Inv0007    P100007              S00007                        18 Avail
able           18-SEP-26

7 rows selected.
```

18 Find inventory with low stock

The query filters inventory records where STOCK_QUANTITY is below the threshold shown, helping identify items that may need replenishment.

```
SQL> SELECT *
  2  FROM INVENTORY
  3  WHERE STOCK_QUANTITY < 10;

INVENTORY_ PRODUCT_ID           SELLER_ID           STOCK_QUANTITY STOC
K_STATUS        LAST_UPDATED
---------- -------------------- -------------------- -------------- ----
--------------- --------------------
Inv0001    P100001              S00001                          4 2026
-09-12          12-SEP-26
Inv0002    P100001              S00002                          4 Avai
lable           13-SEP-26
Inv0005    P100005              S00005                          8 Avai
lable           16-SEP-26
```

19 Join customers, orders, order details, and products

This multi-table JOIN combines customer names with product names through related order tables, demonstrating how matching keys connect records across tables.

```
SQL> SELECT C.C_NAME,
  2         P.PRODUCT_NAME
  3  FROM CUSTOMER C
  4  JOIN ORDERS O
  5      ON C.CUSTOMER_ID = TO_NUMBER(SUBSTR(O.CUSTOMER_ID, 2))
  6  JOIN ORDER_DETAILS OD
  7      ON O.ORDER_ID = OD.ORDER_ID
  8  JOIN PRODUCT P
  9      ON OD.PRODUCT_ID = P.PRODUCT_ID;

C_NAME               PRODUCT_NAME
-------------------- ------------------------
DEVI                  RESIN KEYCHAIN
DIVI                 RESIN BOOKMARK
KAVYA                Resin Pendant
DEVI                 Resin Art Photo Frame
DIVI                 Resin Art Bookmark
```

20 Filter orders within a date range

The query uses TO_DATE to convert date strings and selects orders within the specified date range.

```
SQL> SELECT *
  2  FROM ORDERS
  3  WHERE ORDER_DATE >= TO_DATE('01-08-2026','DD-MM-YYYY')
  4  AND ORDER_DATE < TO_DATE('01-09-2026','DD-MM-YYYY');

ORDER_ID CUSTOMER ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------- -------- --------- ------------ ----------------------
0201     C201     22-AUG-26          750 Pending
0202     C202     23-AUG-26         1000 Shipped
0203     C203     24-AUG-26          500 Delivered
0204     C204     25-AUG-26          750 Pending
0205     C205     26-AUG-26         1000 Shipped
0206     C206     27-AUG-26         1200 Shipped
0207     C207     28-AUG-26          700 Delivered
0208     C208     29-AUG-26         1900 Delivered

8 rows selected.
```

21 Count all customers

COUNT(*) returns the number of rows in CUSTOMER. The displayed result is 8 customers.

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS
  2  FROM CUSTOMER;

TOTAL_CUSTOMERS
---------------
              8
```

22 Display customer record

```
SQL> SELECT *
  2  FROM CUSTOMER
  3  WHERE REG_DATE >= TO_DATE('01-10-2026','DD-MM-YYYY')
  4  AND REG_DATE < TO_DATE('01-11-2026','DD-MM-YYYY');

CUSTOMER_ID C_NAME               EMAIL                                          PHONE_NO ADDRESS
----------- -------------------- ---------------------------------------------- ---------- ---------
REG_DATE
---------
        201 DEVI                 devi@gmail.com                                 9000000001 Chennai
22-OCT-26

        202 DIVI                 divi@gmail.com                                 9000000002 Chengalpattu
22-OCT-26

        203 KAVYA                kavya@gmail.com                                9000000003 Tambaram
22-OCT-26

        101 Arun                 arun@gmail.com                                 9876543210 Chennai
19-OCT-26

        102 Priya                priya@gmail.com                                9876543211 Madurai
19-OCT-26

        103 Kumar                kumar@gmail.com                                9876543212 Coimbatore
19-OCT-26

        104 Divya                divya@gmail.com                                9876543213 Trichy
19-OCT-26

        105 Karthik              karthik@gmail.com                              9876543214 Salem
19-OCT-26


8 rows selected.
```

23 Count completed orders

COUNT(*) with ORDER_STATUS = 'Completed' counts completed orders. The displayed result is 0.

```
SQL> SELECT COUNT(*) AS COMPLETED_ORDERS
  2  FROM ORDERS
  3  WHERE ORDER_STATUS = 'Completed';

COMPLETED_ORDERS
----------------
               0
```

24 Count cancelled orders

COUNT(*) with ORDER_STATUS = 'Cancelled' counts cancelled orders. The displayed result is 0.

```
SQL> SELECT COUNT(*) AS CANCELLED_ORDERS
  2  FROM ORDERS
  3  WHERE ORDER_STATUS = 'Cancelled';

CANCELLED_ORDERS
----------------
               0
```

25 Count pending orders

COUNT(*) with ORDER_STATUS = 'Pending' counts pending orders. The displayed result is 2.

```
SQL> SELECT COUNT(*) AS PENDING_ORDERS
  2  FROM ORDERS
  3  WHERE ORDER_STATUS = 'Pending';

PENDING_ORDERS
--------------
             2
```

26 Find the highest-priced product

The final query uses a subquery to compare product prices with the maximum price and identify the highest-priced product.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE = (SELECT MAX(PRICE) FROM PRODUCT);

PRODUCT_ID PRODUCT_NAME                PRICE CATEGORY            DESCRIPTION
---------- ------------------------ --------- ------------------- ------------------------------
       209 Resin Art Photo Frame        600.00 Resin Art           Customized resin photo frame
```

Result

The SQL statements shown in the supplied demonstrate data retrieval, filtering, sorting, multi-table joins, date-based selection, aggregate counting, and subquery-based price analysis. The displayed outputs include 8 customer records, 0 completed orders, 0 cancelled orders, and 2 pending orders for the respective count queries.